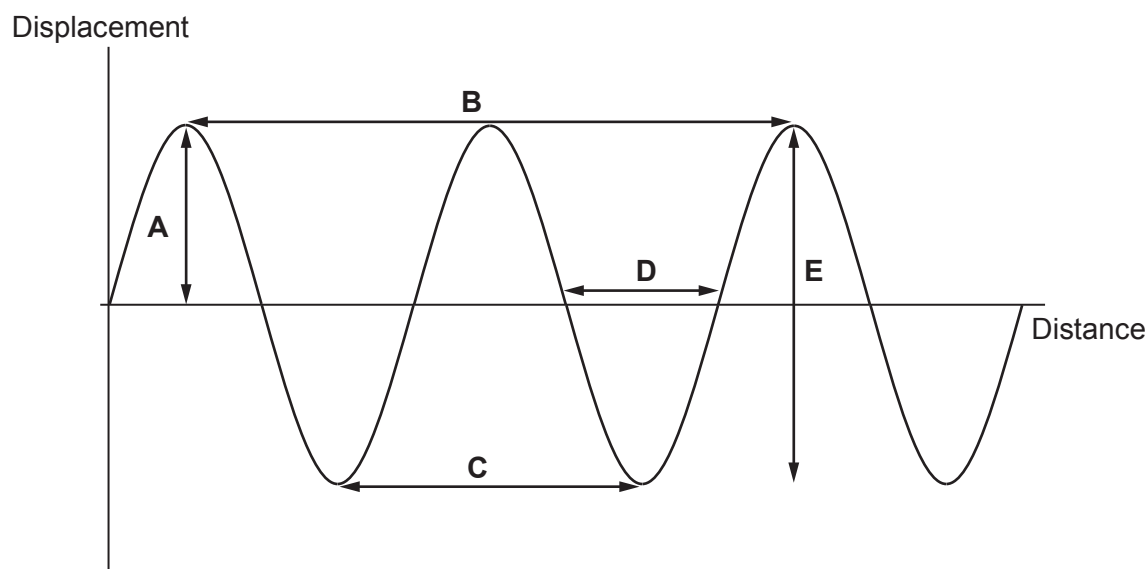


2. The diagram represents water waves on the surface of a swimming pool.



- (a) How many complete waves are shown in the diagram? [1]
- (b) Which letter **A**, **B**, **C**, **D** or **E** shows the amplitude of the wave? [1]
- (c) Select a word from the box to complete the following sentences.

frequency	time	speed	wavelength	amplitude
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- (i) The number of waves per second is the [1]
- (ii) The maximum displacement of the wave is the [1]
- (d) A wave peak takes 0.5s to travel the distance labelled **B** on the diagram. The speed of the wave is 20 cm/s.

- (i) Use the equation:

$$\text{distance} = \text{speed} \times \text{time}$$

to calculate the distance travelled by the wave. [2]

Distance = cm

- (ii) Calculate the wavelength. [1]

Wavelength = cm

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(e) Water waves are transverse waves.

(i) Give another example of a transverse wave.

[1]

.....

(ii) Sound waves are not transverse waves. What type of waves are they?

[1]

.....

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